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E2D-GS

Spec document

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ID Project Name	EYE2DRIVE
Date	25 July 2023
Revision	00
Status	Pending
Approved by	
Number of Pages	24

Scope / Overview	Specification of E2D-GS chip

Last revision	01	Data	08/04/2025

Record of revisions	Rev Data	Rev n°	Notes
	27/01/2025	00	First release
	08/04/2025	01	Typing revision Added load column in Tab. 5.2 Other info about pins' load in par.5.1.6 Info about the output buffer added in par. 5.1.2 Updated Fig. 6.1 Fig. 4.1 updated with package numbering Package mechanical outline added in par. 4.2 Updated pin list in par. 4.1

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1. Introduction

This technical document aims to introduce the **E2D-GS** chip, a prototype designed to showcase the robust performance of EYE-TECH imaging technology in uncontrolled light environments. It is essential to emphasize that this chip is not intended for production; rather, it serves as a crucial step to address time, cost, and priority considerations. The pressing need for a prototype, coupled with the opportunity to access free development tools for three months from mid-August onwards, has steered us towards this approach.

The primary objective behind the **E2D-GS** chip is to deliver a sensor with a reasonable resolution suitable for applications' PoCs, while significantly simplifying the architecture to mitigate implementation risks and expedite the design process. The critical enhancement over the existing ET1080 sensor is the integration of a global shutter, enabling effective high dynamic range (HDR) performance even in unpredictable lighting conditions.

The **E2D-GS** chip is envisioned as a monolithic camera-on-a-chip sensor, with a resolution of 800x600. Colour filters are added on top as well as microlenses. For configuring the sensor, manual adjustments will be made using a series of digital pins for array addressing and output data management (22 signals). Reference signals (10 signals) will be provided to the chip as analogue inputs.

The sensor is produced in Tower Semiconductor, 180nm front-side CIS process. Throughout this document, we will provide detailed insights into the **E2D-GS** chip's specifications, features, and design considerations.

2. Key Features

- 800x600-pixel resolution,
- 4.8um pixel pitch with global shutter
- 4mmx3mm sensitive area
- Tuneable dynamic range through patented technology
- Adjustable sensitivity
- Light source control signals
- 10MHz data rate
- Double analog outputs,
- 3.3V operating voltage

3. Architecture specification

The architecture has been decided keeping in mind that the sensor is a test chip focused to demonstrate the new approach proposed in the last patent. For this reason the architecture is minimal to decrease implementation risks.

As shown in Fig. 3.1, the main blocks of the sensor are the pixel array, the decoders to each address the single pixel and the readout chain, made by the Correlated Double Sample (CDS) blocks (one for each column of the array) and a buffer to carry signals outside the sensor.

The control logic of the sensor will NOT be integrated in the sensor itself but provided externally by the control signals (22 signals). Also, the input reference voltages and currents needed for the sensor operation will be generated externally (10 signals), including the reference ramp that gives rise to the HDR behaviour. High flexibility in the pixel control is mandatory to manage and later optimize global shutter management. For this reason 3 row and 2 column decoders have been integrated.

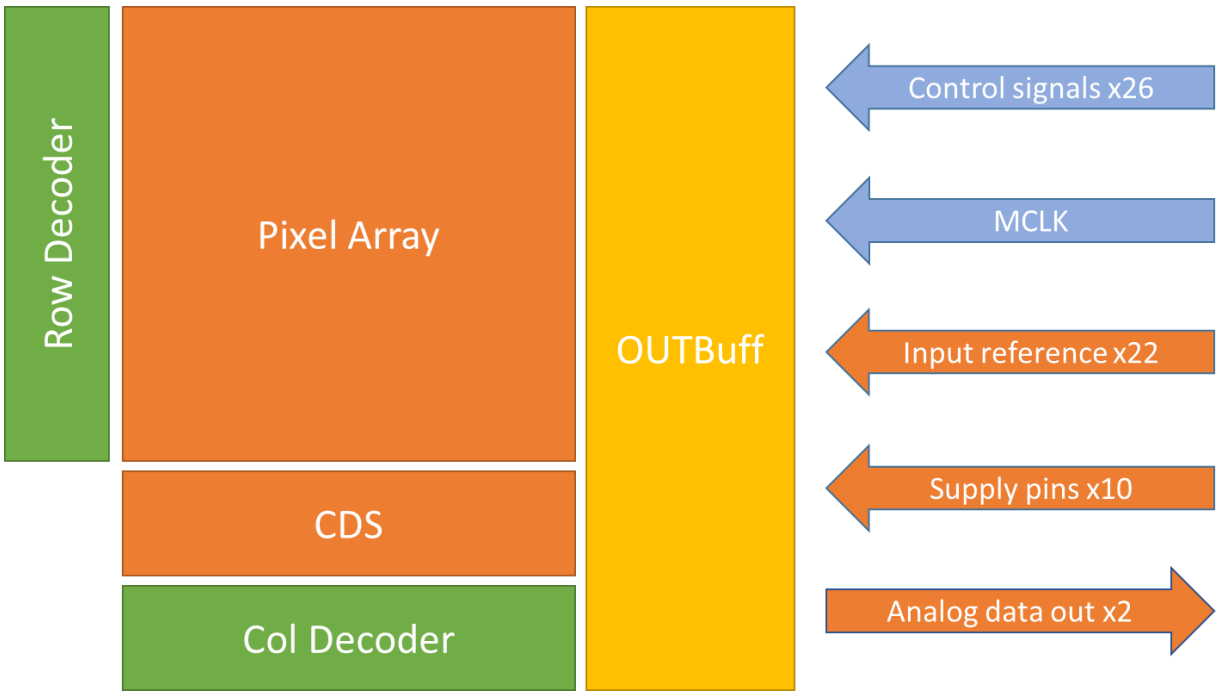


Fig. 3.1 Basic sensor architecture

4. Pin Configuration

4.1. Pin list

The E2D-GS sensor pin list is reported in the following table:

Pin	Pin name	Dir	Type	Description
DIGITAL				
1	RowTx2_FT0			Row Tx2 global mask to 0 signal
2	SampleCDS	I	D	CDSs input enable for data and reset sampling
3	VDDO1	S	P-P	3.3V PAD Ring power supply
4	VSSO1	S	P-P	0.0V PAD Ring power supply
5	RowTx2_D	I	D	Row Tx2 decoder data input
6	NC			
7	RowSEL_D	O	D	Row SELECT decoder data input
8	RowRES_D	I	D	Row RESET decoder data input
9	ColSEL_D	I	D	Column SELECT decoder data input
10	ColRES_D	I	D	Column RESET decoder data input
11	DecCLK	I	D	Decoder clock
12	ColRES_FT0	I	D	Column RESET global mask to 0 signal
13	ColRES_S0	I	D	Column RESET global relet to 0
14	ColSEL_FT0	I	D	Column SELECT global mask to 0 signal
15	ColSEL_S0	I	D	Column SELELCT global relet to 0
16	ColRES_Q	O	D	Column RESET data output signal
17	ColSEL_Q	O	D	Column SELECT data output signal
18	NC			
19	NC			
20	VbCDS	I	A	CDS reference voltage
21	NC			
22	En_OUT	I	D	OUTPUT buffers enable
23	NC			
24	VSSO2	S	P-P	0.0V PAD Ring power supply
25	VDDO2	S	P-P	3.3V PAD Ring power supply
26	IBIasCDS	I	A	CDSs' polarization current sink
27	IBiasCol	I	A	IBiasCol polarization current source
28	Data_OUT1	O	A	Pair columns' data output
29	Data_OUT0	O	A	Odd columns' data output
30	Iref_Out2	I	A	Odd columns' polarization current source
31	NC			
32	Iref_Out1		A	Pair columns' polarization current source
33	PixSupp	S	P-A	Pixel Array supply voltage (3.0V)
34	Tx1	I	D	Global shutter signal (-0.5V/3.5V)
35	GRST	S	P-A	Global blooming and dynamic control (discharge ramp 0.0V-3.3V)
36	GND	S	P-D	Global ground
37	VDD	S	P-D	Global power supply
38	VDDH	S	P-A	3.5V power supply used as high level for RES and SEL row signals
39	VSSH	S	P-A	-0.5V low level power supply used for low level Tx2 signal

40	RowRES_Q	O	D	Row RESET decoder data output signal
41	RowSEL_Q	O	D	Row SELECT decoder data output signal
42	RowTx2_Q	O	D	Row Tx2 decoder data output signal
43	RowSEL_S0	I		Row SELECT global relet to 0
44	RowSEL_FT0	I		Row SELECT global mask to 0 signal
45	RowRES_S0	I	D	Row RESET global relet to 0
46	RowRES_FT0	I	D	Row RESET global mask to 0 signal
47	RowTx2_S0	I	D	Row Tx2 global relet to 0
48	NC			

Key:	I	Input	D	Digital Pin with programmable drive strength
	O	Output	A	Analog Pin
	I/O	Input / Output	P-P	Power supply for pad output buffers
	S	Power Supply	P-D	Power supply for core logic
			P-A	Power supply for analog core

All ground and power supply pins must be connected to the proper value and not left floating.

Sinking current flows from VDDA through the bias generator to the pin.

Sourcing current flows from the pin through the bias generator to GNDA.

4.1.1. PinOut

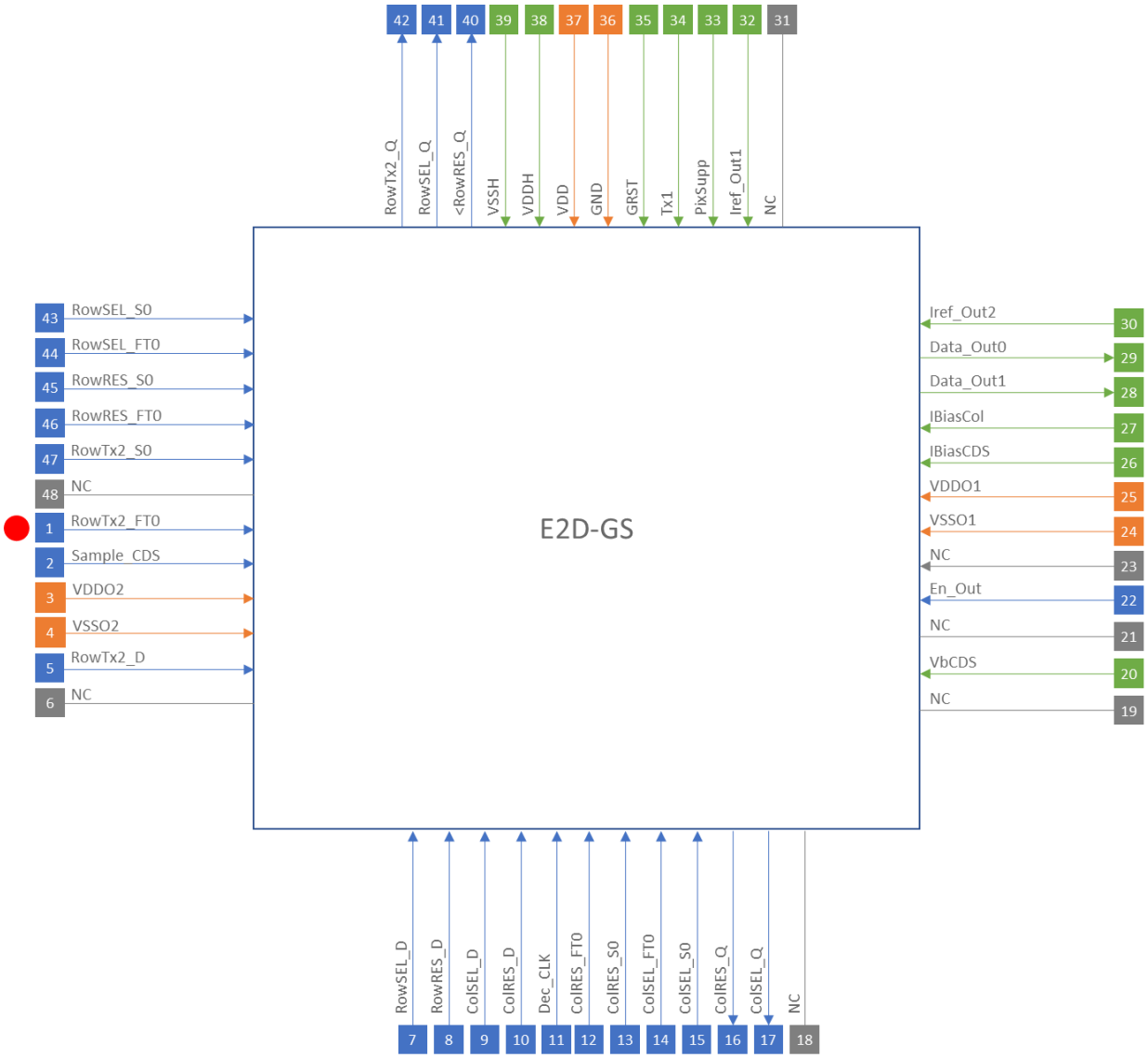
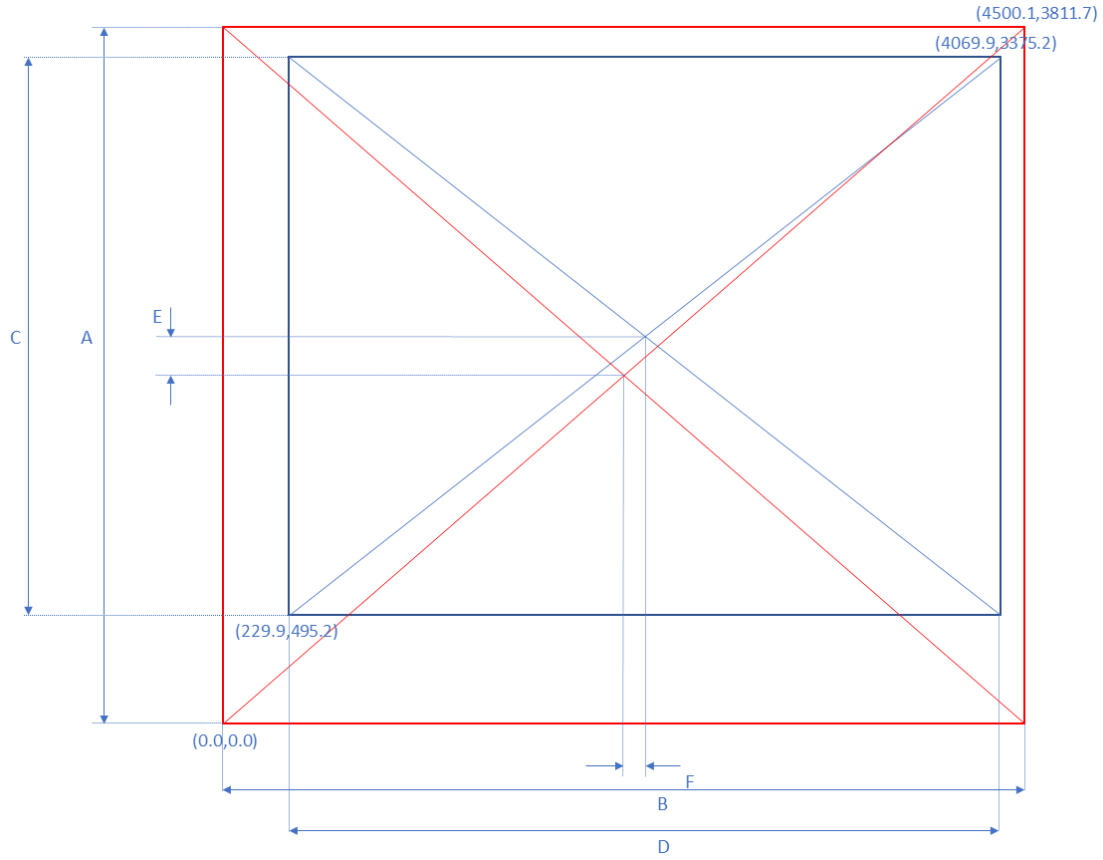


Fig. 4.1 Chip pinout

4.1.2. Die Outline Dimensions and Mechanical Tolerances



Reference	Note	Typ (µm)
A	Y-size of the die	4420.1
X	X-size of the die	5135.1
C	Y-size of the sensor array	3840.0
D	X-size of the sensor array	2880.0
E	Y-offset between sensor array center and die center	465.9
F	X-offset between sensor array center and die center	330.0
	Die thickness	297.0

4.2. Package mechanical outline

SSM P/N LCC04811

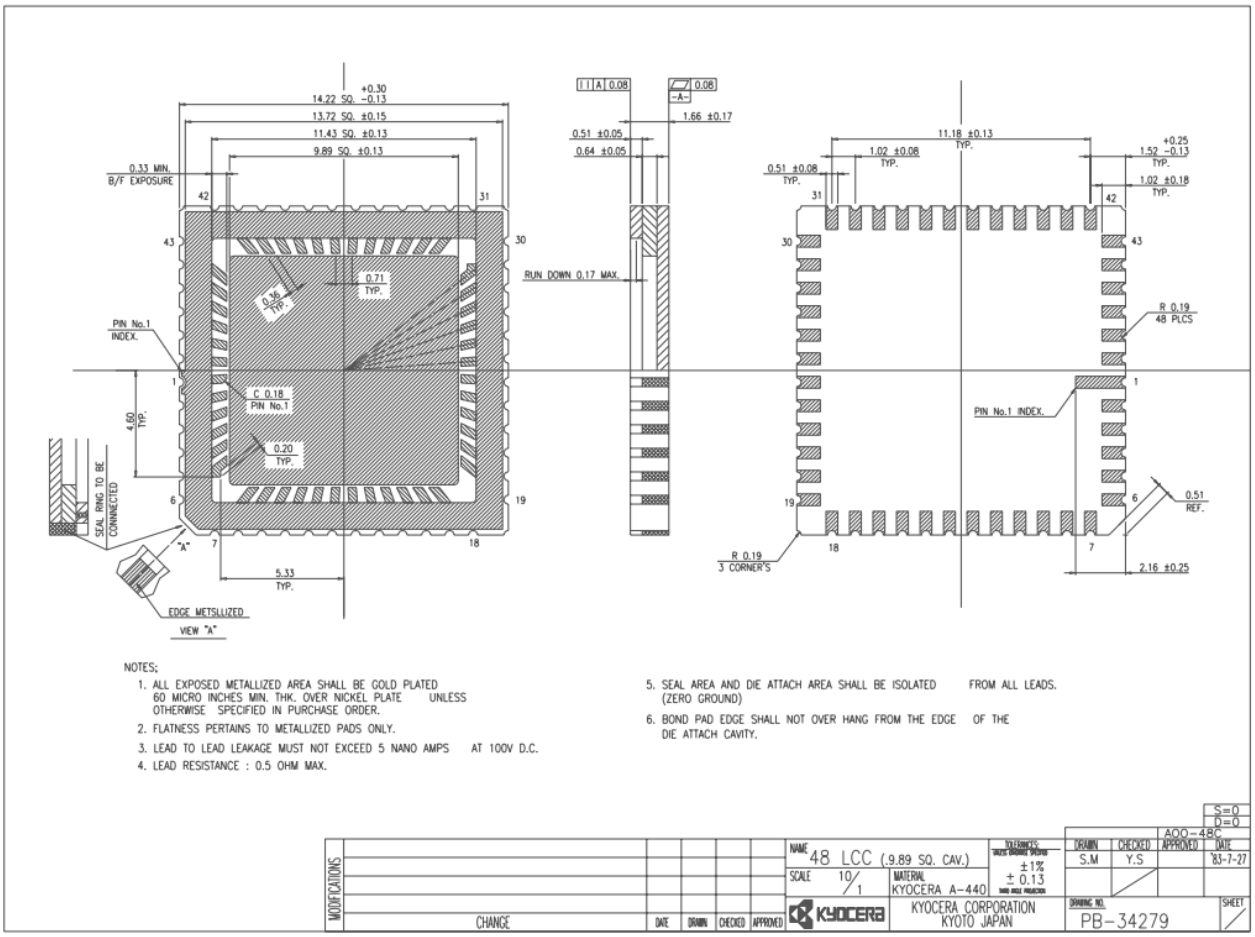


Fig. 4.2 LCC04811 mechanical case outline

5. Operation and timing of the Camera

All functions of the chip are controlled externally in order to maximize the flexibility as in the intention of this functional test chip.

After a complete reset of the array's photodiodes, through a pulse of the **GRST signal**, the whole array of pixels enters the integration phase for a period of time set by the user: the integration time.

During the integration period, photodiodes' discharge is conditioned by the piece-wise-linear signal applied to the gate GRST, global to all the pixels. This signal enables a controlled discharge of the pixels of the array resulting in an output with an increased dynamic, depending on the shape of the conditioning signal.

At the end of the integration the signals at each PD junction are globally transferred to the SN junction, through a **Tx1** global pulse, and ready for readout.

The pixels' signals are transferred row by row to the CDSs and ready for external readout. Before signal reading by enabling Tx2, each pixel output stage of the selected row is reset through a pulse of RST signal and the reset signals transferred to each column's CDS for sampling. After that, RST is released and Tx2 of the selected row enabled for data transfer to the output stage. The signal is now ready to be sampled by the CDS that, in the meantime, switched the ResCDS signal from high to low to enable the column double sampling and signal subtraction. Tx2, RST and ResCDS are, respectively, the output of the decoders RowTx2, RowRES and ColRES, Details about the decoders and the timing to generate the proper signal sequence, are reported in sections 5.1.4 and 7.

After the freezing of the signal at CDS level, the columns are read by two readout channels one after the other (Pair and Odd by two different channels in parallel). The interface between the CDSs and the output is a buffer (one for each of the two channers).

For each pixel a double readout must be performed, the signal first and the reset after, both synched with the CDS selection ad reset.

This behaviour is summarized in Fig. 5.1 and Fig. 5.2 while in section 5.1.1 are summarized more details about the pixel and in 5.1.4 about decoders behavior.

It is worth underlining that global shutter operation allow parallel readout and integration and a new integration can start in parallel to readout as soon as Tx1 is again closed after signal transfer.

A flow chart of the operation of the sensor is reported in Fig. 5.3.

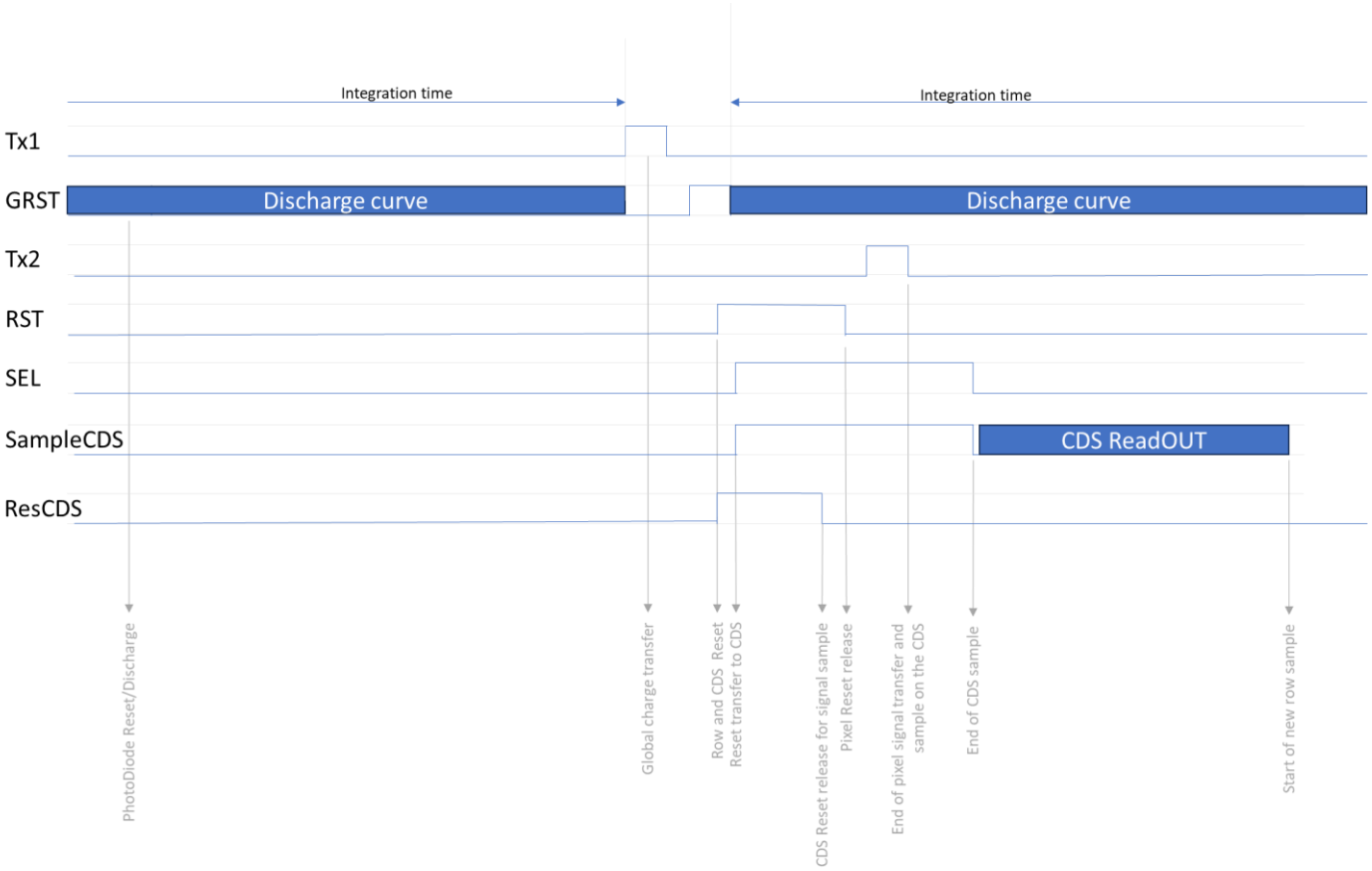


Fig. 5.1Integration and pixel sampling

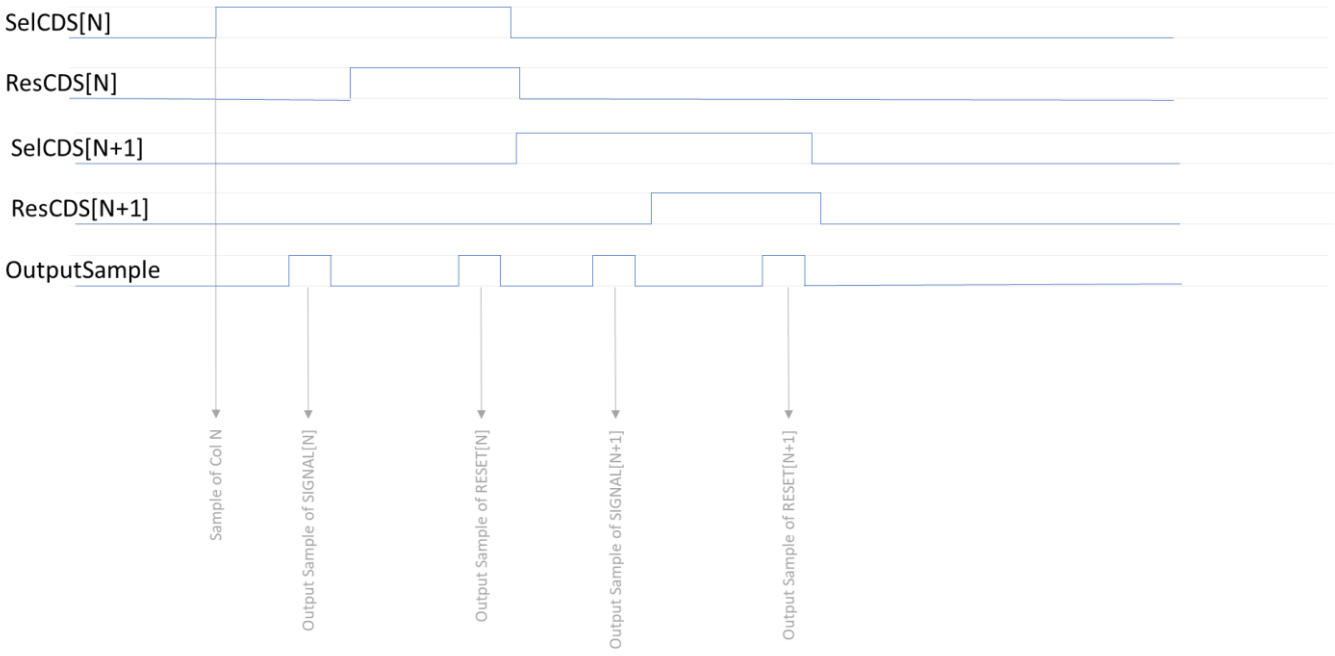


Fig. 5.2 Columns' readout

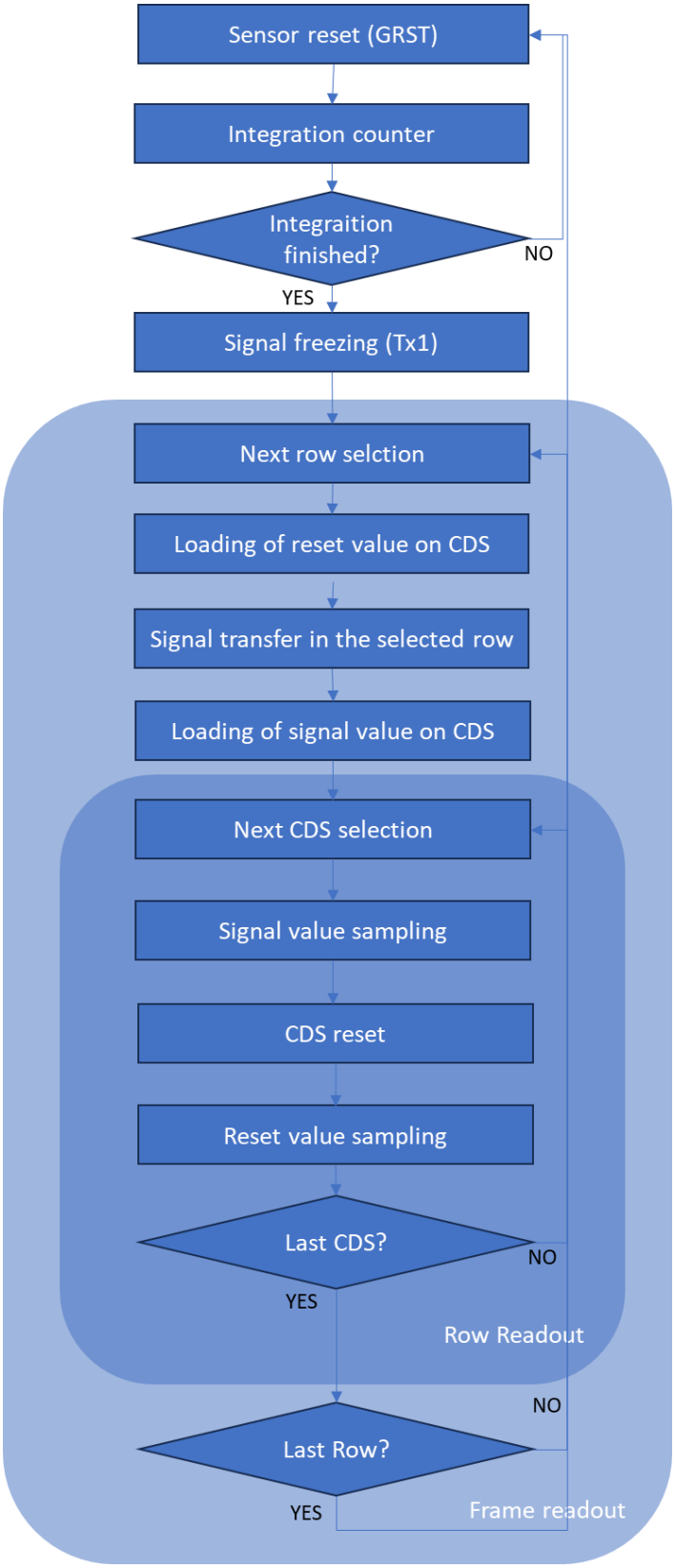
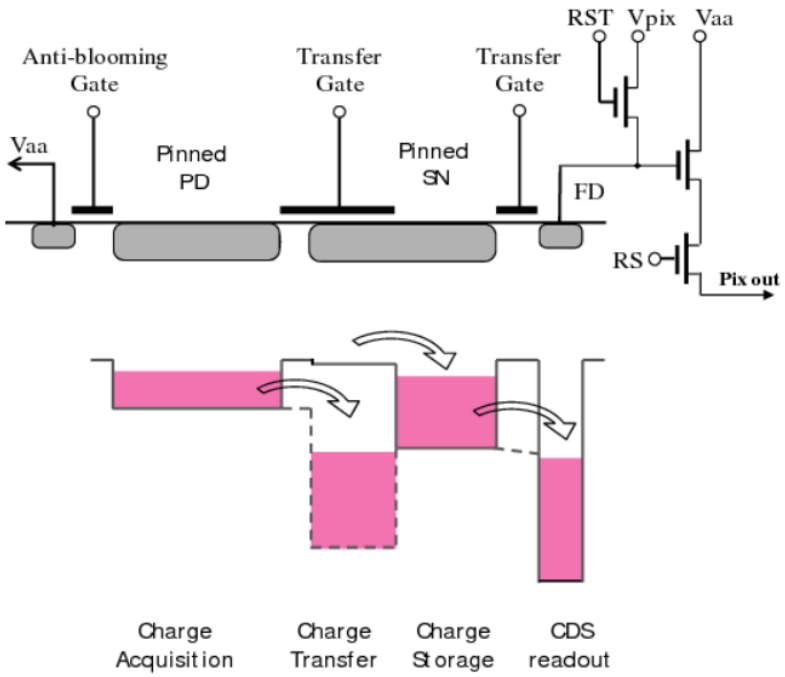


Fig. 5.3 Flow diagram describing the operations of the sensor.

5.1.1. Pixel Array



Supply	Level
Vdd Pix	3V
GRST Hi	3.3V
GRST Low	0V
RST Hi	3.5V
RST Low	0V
SEL Hi	3.5V
SEL Low	0V
TX1 Hi	3.5V
TX1 Low	-0.5V
TX2 Hi	3.3V
TX2 Low	-0.5V

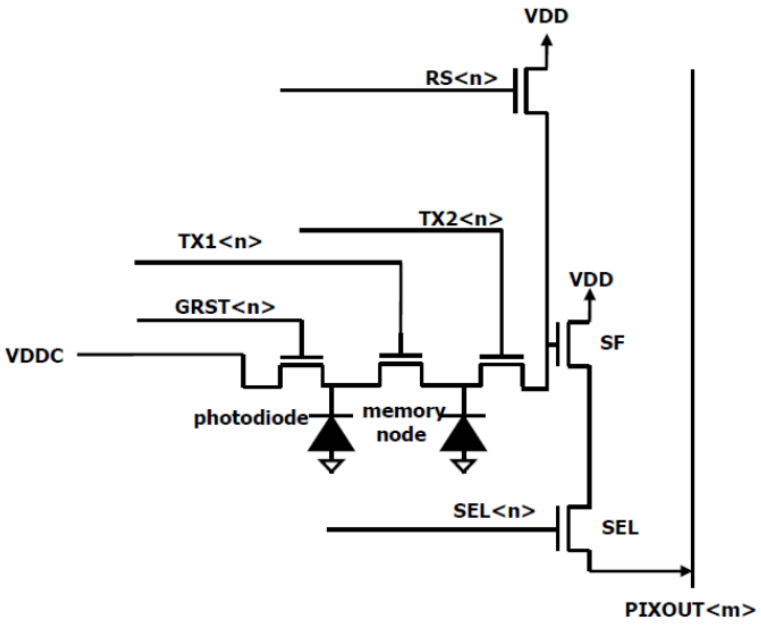


Fig. 5.4 Pixel configuration.

The array of pixels is made by 800x600 square pixels with a pitch of 4.8um. Each pixel contains a photodiode that converts the incoming light into a continuous voltage. The pixel architecture is a standard pinned photodiode with flush/shutter gate and global shutter (3-transistor, 1 transfer gate, 1 shutter gate and 1 photodiode, Fig. 5.4). High dynamic range (HDR) is achieved by conditioning the shutter gate with a step-wise signal, applied to the GRST terminal, externally generated.

The pixel configuration guarantees a programmable dynamic range from 65 dB to over 100 dB by changing the shape of the conditioning signal VRES.

Pixel type	Global shutter 6T CDS
Technology	CIS 180nm FSI
Conversion gain	110 uV/e-
Full-well	8.5 ke-
Native dynamic range	65dB
Noise	4e-

Tab. 5.1 Pixel characteristics

5.1.1.1. Conditioning signal GRST

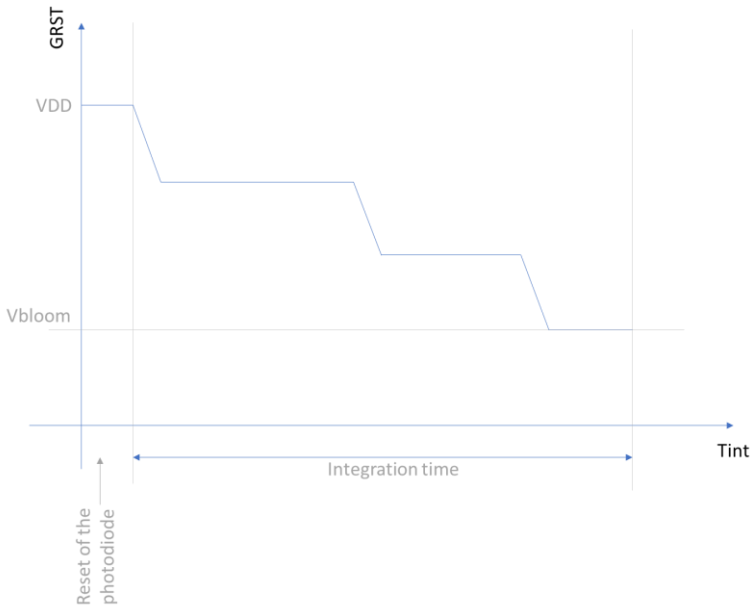


Fig. 5.5 GRST ramp generation shape.

GRST is a special analog reference voltage for the pixel array. GRST is changed during the integration time according to a piece-wise-linear non-increasing curve. At the end of the integration time the photodiode signal is freezed by a transfer to the floating gate (FD) and ready for readout. Because of physical decoupiling of the photodiode (PD) junction from the readout of the pixel, another integration and another conditioning curve can start in parallel to readout.

The GRST is generated externally, and a possible shape is shown in Fig. 5.5. The only constraint for the ramp generator is that the GRST value cannot be increasing, therefore there can be plateaus.

The integration time can be between 1msec and 100msec with a typical value of 30msec.

5.1.2. Video Line

When a row-decoder selects a row, every pixel of the selected row transfers simultaneously its output value to the corresponding column CDS which stores the reset value. Then each pixels of the selected row transfer its signal from FD to the output and their signal values are transferred to the corresponding CDS. In this way the two samples are correlated. The CDS subtract the reset value from the stored signal and makes the result available at the output. Such subtraction is a standard technique to mitigate *Fixed Pattern Noise (FPN)* (see section 5.1.3 for more details). Since the correlation between the two samples, also Pixel temporal noise is subtracted.

The CDSs are then sequentially selected in groups of 2 by a column decoder and the output values are transferred to 2 output buffers, one for each selected column.

The buffer output is in low impedance. A 0Ohm series resistance is suggested on the output for fine tuning during test. The output bugger can drive a capacitance up to 20pF.

A double sampling is required by the external ADC. The first one, during the first phase, is the signal value, the second, during the coming phase, is the reset value. The two samples must be subtracted externally after sampling to subtract the column FPN introduced by the CDSs.

5.1.3. Fixed Pattern Noise

Fixed Pattern Noise (FPN) is a time-invariant noise in images. The FPN is caused by the non-uniformity of the transistor's characteristics within repetitive devices like pixels and column amplifiers, resulting from fabrication process tolerances. Threshold voltage variations introduce a voltage-offset characteristic of each pixel and gain variations introduce a gain error. As described above, pixel FPN offset subtraction is performed inside the sensor by CDS blocks while column FPN generated by the CDSs must be subtracted externally by double sampling the signals, first during the signal phase and after during the reset phase. However, it is recommended to perform also a subtraction using software-based techniques.

5.1.4. Decoders

Decoder are used to address the pixels of the array. Five decoders are used inside the sensor:

- one for row selection (RowSEL);
- one for row reset (RowRES);
- one for row signal transfer (RowTx2)
- one for column selection (ColSEL);
- one for column reset (ColRES).

. The decoders are based on a shift register architecture with access to the data I/O of the first and last cells only. Each decoder can be synchronously set to 0 (by the SEL_S0 signal and decoders enable, SEL_EN), while the output signal going to the pixel array can be asynchronously forced to 0 by the SEL_FT0 signal without affecting the logical value stored inside. The decoder acts on the rising edge of the decoders signal enable, SEL. A schematic view of a decoder cell is shown in Fig. 5.6.

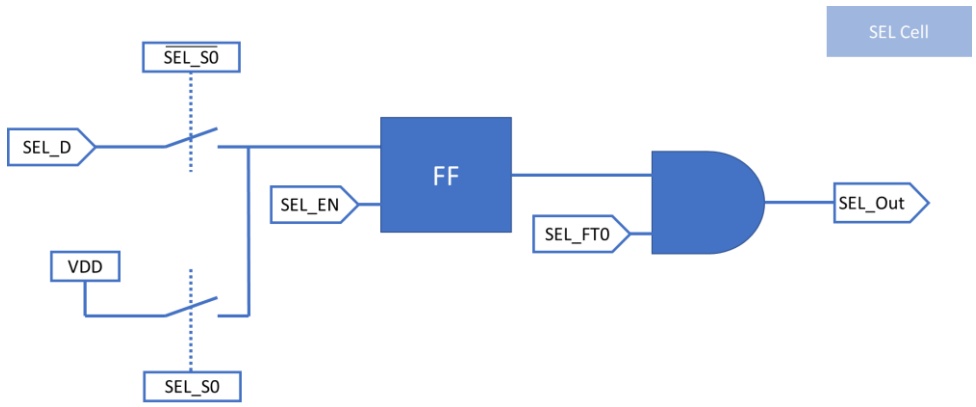


Fig. 5.6 Schematic diagram of a decoder cell

5.1.5. CDS

The **CDS** block is crucial for reducing the noise that arises from variations in transistor parameters within the pixels and readout circuits. Each column of the CMOS sensor is equipped with a CDS, which sequentially reads the pixel's output signal and its reset value. The CDS in this design is based on a simple switched capacitor architecture and uses a source follower for decoupling. This configuration allows for isolation and buffering of the signal from the pixel before performing the subtraction between the light signal and the reset signal. This subtraction compensates for the noise introduced by variations in the transistor characteristics within the column and between pixels, thereby improving the quality of the acquired signal.

The device will work on switching signals. Sample timing is not critical since sample is done ones a line. Readout is more time sensing, maximum expected sampling rate is 20MHz.

5.1.6. Analog references

The needed analogue references are:

- 4 current references: IBiasCDS, IBiasCOL, IrefOUT1, IrefOUT2.
- 2 voltage references: VbCDS, GRST. All of them are externally generated.

Typical values are reported in the following tables

	Description	Typ.Value (µA)	Load
IBiasCOL	Bias current for bit-lines	5	200Ohm
IBiasCDS	Bias current for column CDSs	5	200Ohm
IrefOUT1	Bias current for signal output 1 (Odd)	5	200Ohm
IrefOUT2	Bias current for signal output 1 (Pair)	5	200Ohm

	Description	Typ. Value (mV)	Load
VbCDS	Bias voltage for column CDSs	1.5V	20uA x 800
GRST	Global blooming and dynamic control	Dynamically Programmable TODO: RANGE?	Low current load (800x600 minimum gates)

Tab. 5.2 Typical values for current (above) and voltage (below) references.

6. Integration reference

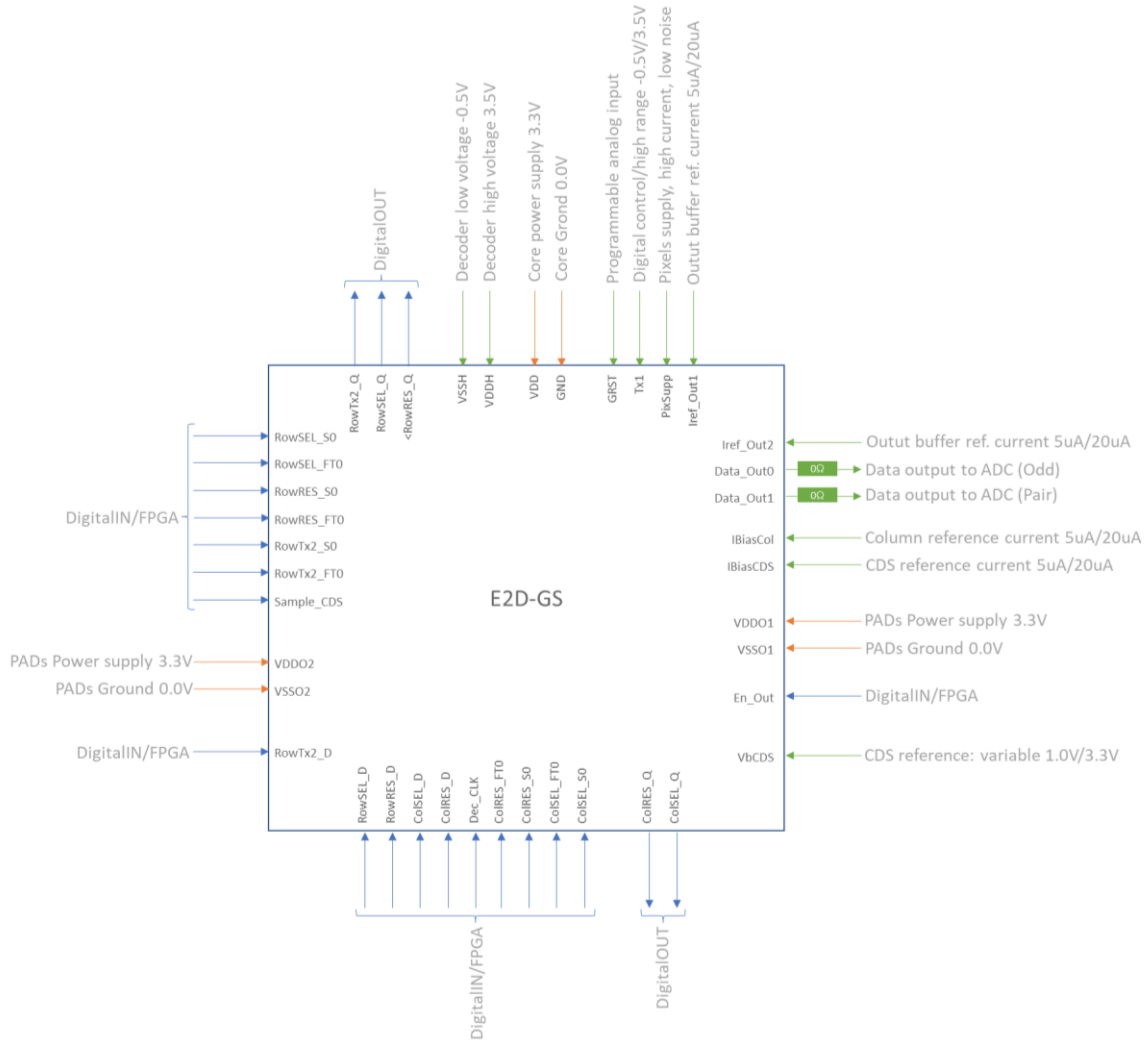


Fig. 6.1 E2D-GS externa connections

7. Appendix A – data timing

In this section timing diagram to control E2D-GS are included.

7.1. Integration/Readout Phase of the E2D-GS Sensor

The first stage of the **E2D-GS** sensor control concerns light signal acquisition. Like all integration-based sensors, the signal is captured during a defined integration period, during which the photodiode is left free to react to the incident light.

In the specific case of the pixel integrated into the **E2D-GS** chip, the pixel consists of a pinned photodiode with global shutter. The key signals involved in the acquisition process, and synchronous to the master clock f the system (MCLK), are:

- GRST (Global Reset): This signal is typically used for antiblooming and for resetting the photosensitive area before starting the signal acquisition. In the case of the E2D-GS sensor, GRST is dynamically controlled, as detailed in Section 5.1.1.1.
- Tx1 (Transfer Gate): This signal is responsible for transferring the charge from the photodiode to the floating diffusion (or memory node, see Section 5.1.1 related to the pixel structure). The charge transfer occurs simultaneously for all pixels, moving the stored signal from the photodiode to the floating diffusion, where it is held until readout.

Once the charge transfer is completed, the frame readout can begin. At the same time, a new integration cycle can start in parallel.

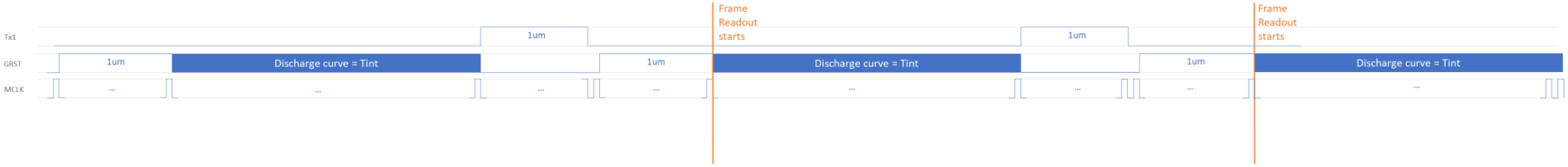


Fig. 7.1Integration and readout phases

It is crucial to ensure proper synchronization between the integration and readout phases, even when their durations do not match. Specifically, two scenarios must be considered:

- When the integration time is shorter than the readout time.
- When the integration time exceeds the readout time.

These cases require careful management to avoid artifacts and ensure optimal signal acquisition and processing.

Errore. L'origine riferimento non è stata trovata. illustrates the row addressing mechanism and the transfer of reset and signal control to the pixels within the selected row. The row decoder, selection, reset, and Tx2 signals are all activated simultaneously but are made available to the pixels in a sequential manner, depending on the state of the xx_FT0 signals.

For each row, the process begins with resetting the entire decoder using a pulse on the xx_S0 signals. It is important to note that S0 is synchronous and requires a Dec_Clk pulse to become active, whereas xx_FT0 is asynchronous and operates independently of Dec_Clk.

Following the initial reset, the first row is selected by driving the xx_D signals high for one clock cycle. The first Dec_Clk pulse corresponds to the selection of the first row. For subsequent row selections, a number of clock pulses equal to the row index is applied. It is crucial that xx_D remains high only during the first clock pulse of the selection sequence.

Once a row is selected, the first signal propagated to the pixels is the reset signal (RST), which resets the pixel's output stage. This reset signal is simultaneously transferred to the next stage of the readout path (Correlated Double Sampling - CDS) while activating the selection signal (SEL). The SEL signal connects the selected row's pixels to the output bus, while the Sample_CDS signal enables the CDS to capture the incoming signal.

After a sampling period of 500 microseconds, RES_FT0 is driven high, leading to the deactivation of the RST signal. Immediately afterward, Tx2 is activated by pulling Tx2_FT0 low, allowing the integrated signal to transfer from the floating diffusion (or memory node) to the output node, making it available for readout. This marks the beginning of signal transfer to the CDS, where the reset signal is subtracted from the pixel signal.

Following this step, the CDS stages are disconnected from the row and sequentially discharged, completing the readout cycle.

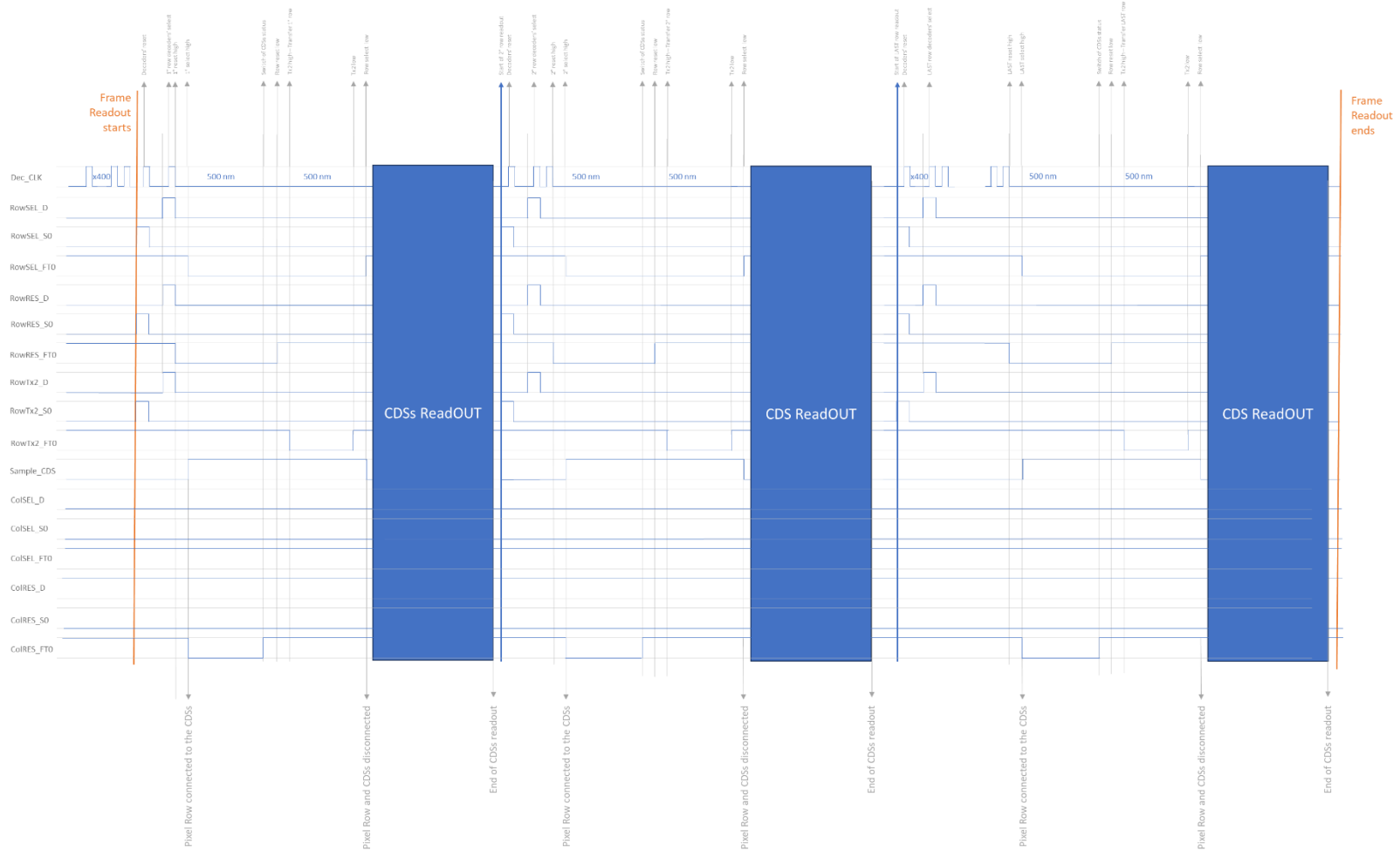


Fig. 7.2 Frame readout – row selection

During the phase in which Sample_CDS is active and the pixels are being sampled, the CDS stage operates in two distinct phases.

- First phase (Reset Sampling): The reset signal of the pixel is sampled, during which the reset signal of all CDS stages remains high.
- Second phase (Signal Sampling): The pixels transmit the acquired signal, and during this phase, the reset signal is deactivated.

This process enables the CDS to perform a subtraction between the two sampled values, effectively removing the fixed pattern noise. The resulting signal is then stored for subsequent readout.

As shown in Fig. 7.3, at the beginning of the CDS readout phase, all CDS units have stored the pixel-acquired signal within their internal capacitance. At this stage, no CDS is selected, and all reset signals are deactivated since ColSEL_FT0 and ColRES_FT0 are both held high.

The selection and reset decoders are first reset to 0 before initiating the readout phase. Immediately after, the ColSEL_D signal is driven high for one clock cycle, while the ColRES_D signal is set high and remains high throughout the CDS readout phase, ensuring that the decoder is fully loaded with logic 1 in preparation for the next phase.

During readout, the CDS units are sequentially selected and reset, transferring first the signal value and then the reset value to the output. These two values are then externally retrieved in sequence and subtracted outside the chip to obtain the final corrected signal.

To perform the selection and reset operations, a Dec_Clk pulse is applied. Then, ColSEL_FT0 and ColRES_FT0 are sequentially driven low, ensuring that:

- The selected CDS unit is connected to the output, allowing the signal value to be read.
- The reset operation is performed immediately after selection, making the reset value available for readout.
- This process ensures that each CDS is correctly read, reset, and prepared for the next acquisition cycle.

Signal and reset are read in sequence and subtracted externally to the chip.

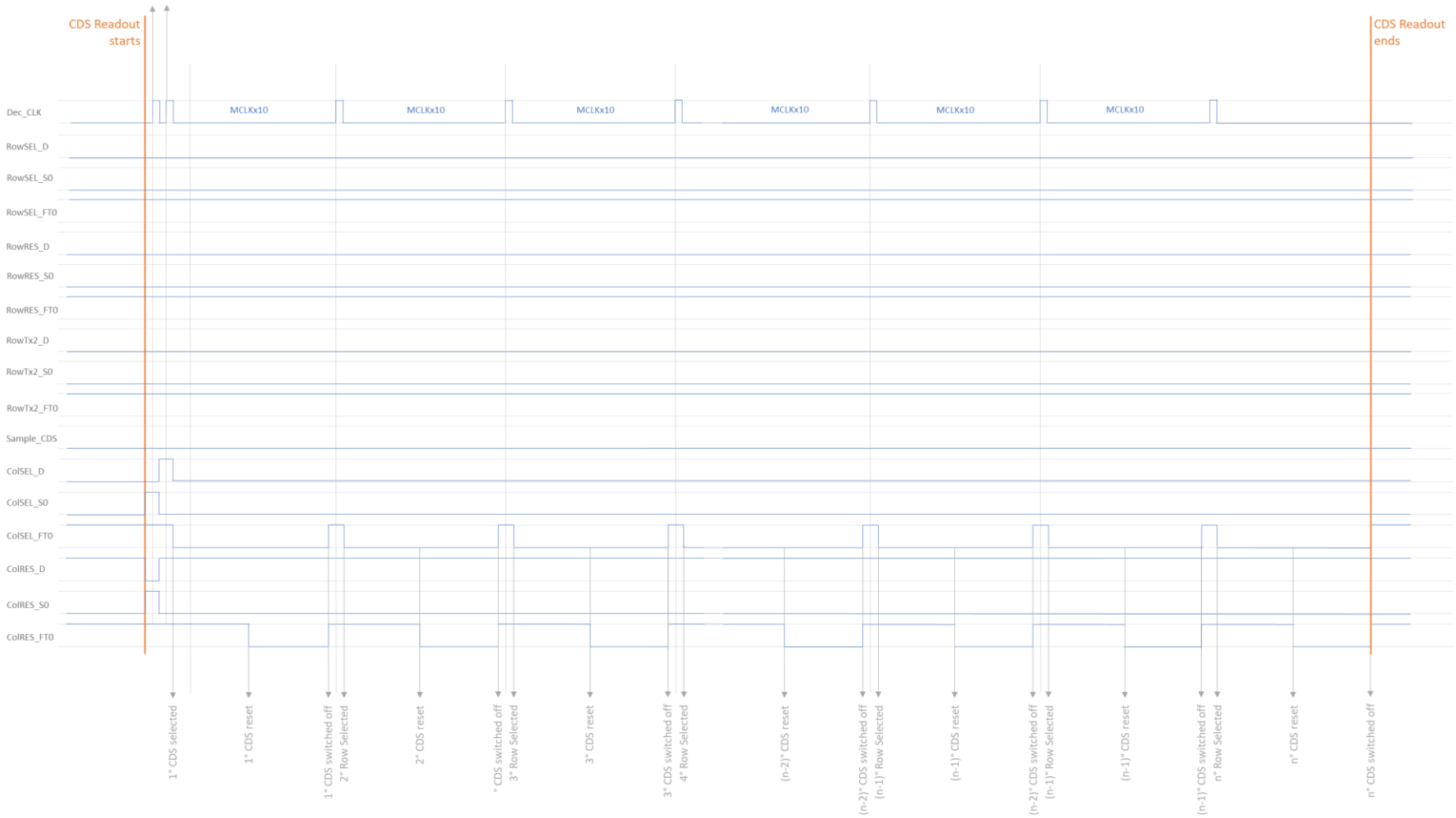


Fig. 7.3 CDSs' readout